DOD-STD-1312
19 October 1983

MILITARY STANDARD

FASTENER TEST METHODS, METRIC



FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

FASTENER TEST METHODS, METRIC

DOD-STD-1312

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specification and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. Background. The selection of fasteners in the design and construction phase of weapons systems and equipments and the analysis of fasteners used therein for parameters, such as suitability, standardization, and simplification requires the evaluation and correlation of test data and criteria. In evaluating these functions, it has become increasingly evident that the data and criteria received from prime contractors, fastener producers, private laboratories, military documents, and military laboratories cannot be adequately correlated. The chief reason for this is the absence of unified standards prescribing test methods, procedures, fixtures, data analysis, and data presentation. This disparity encourages and often requires duplication of testing and reduces reliability of weapons systems and equipments, thereby impairing operations involving engineering control, design maintenance, quality assurance, standardization, and logistics simplification. To correct this situation it was established that a unified set of test methods and procedures is required to serve as a basic engineering tool for obtaining and presenting engineering data. As a result, this document was developed and prepared to satisfy this vital need. This standard delineates standard test methods, procedures, conditions, fixtures, and methods of recording, analyzing and presenting test data. The purpose of the resulting test data is to determine the capability of fasteners to withstand the various environmental and mechanical conditions encountered when used in military weapons systems.

2. Benefits. Many benefits are to be expected by unified standard tests for fasteners. The following is a list of benefits that can be expected to accrue from using this standard.

a. Availability of testing techniques and criteria necessary for the design and development of increasingly superior structural fasteners. In the past, when a new fastener was needed to solve a problem, the necessary test procedures had to be developed and written; also, the necessary fixtures had to be designed. This always imposed delays, additional expenses, variations in test methods and in test results.

b. Increasing incidences of correlation among fastener test data and design engineering techniques and design criteria for weapons systems and equipments. As new systems are designed to achieve higher performance for a given volume and mass of material, it is essential that design data for structural joints (a large portion of which is obtained by testing fasteners) maintain a close correlation with engineering techniques and design criteria. Where weight is of importance, such as in aircraft structures, such close correlation will provide a substantial reduction in weight.

c. Availability of testing techniques and criteria necessary to maintain consistent quality in the procurement of fasteners.

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d. Reduction in the time and effort required for the development of procurement specifications for fasteners by having a series of standard tests available.

e. Elimination of duplication in the development of test methods and documents by individual fastener producers, Government contractors, and military activities.

f. Elimination of duplication of effort in research and evaluation due to lack of unified standard test methods. When test methods are not standard, results cannot be correlated; therefore, producers, prime contractors, and military activities will not accept each other's results and will duplicate research, development, and evaluation work.

g. For logistics simplification, fastener attributes must be reduced to a common denominator so that variation in type can be compared. Data and the presentation thereof, by means of this standard, will be provided in such form that various types and variations in types of fasteners may be compared to determine their degree of interchangeability, substitutability, and replaceability.

h. Time and effort required to reach agreement in regard to specific engineering changes will be reduced by providing unified standard tests for use in evaluating new fasteners.

i. This standard will provide a means for establishing guidelines for evaluating fastener test data and engineering techniques.

j. Industry and Government design personnel will become thoroughly indoctrinated in the use of a single standard prescribing unified standard test methods.

ACKNOWLEDGEMENT

This standard was prepared by the Fastener Testing Development Group (FTDG), consisting of representatives of Government agencies, engineering societies, trade associations, independent testing laboratories, the using industry, and the fastener producing industries.

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1. SCOPE

1.1 Applicability. This standard establishes methods for testing metric fasteners to determine their physical and mechanical properties. These standard test methods will yield reproducible data for use in research, development, procurement or product application. These test methods shall be observed when specifically referenced in the applicable documents. In case of conflict between the provisions of these methods and the individual test procedure in the specification for a particular item, the latter shall take precedence. This document exists as a dynamic standard subject to addition and change as reflected by the "state-of-the-art."

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-962	Outline of Forms and Instructions for the Preparation of Military Standards and Military Handbooks

2.1.2 Documents applicable to a particular test method bookform standard part or section shall be listed therein.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

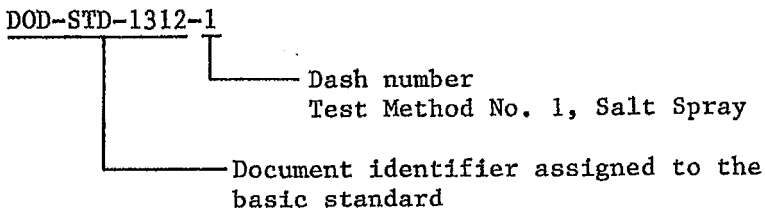
3.1 Definitions applicable to a particular test method bookform standard part or section shall be listed therein.

4. GENERAL REQUIREMENTS

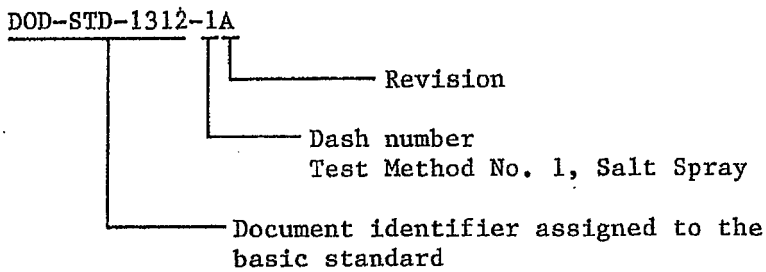
4.1 Test method numbering system. This standard consists of a general basic document and supporting parts or sections. Each part addresses a specific test method which is designated by a dash number commencing with number 1. A different dash number is assigned to each test method.

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4.2 Document format structure. Each test method is prepared as an individual document in the format of a bookform standard in accordance with the guidelines of MIL-STD-962. The individual bookform standard part is identified by the basic document identifier, DOD-STD-1312, followed by a sequential dash number. The dash number is identical to the number of the test method. Example:



4.3 Method of revision. The individual bookform standard part is revised and issued independent of the other parts. Revisions are identified by a capital Gothic letter immediately following the document identifier dash number. Example:



Changes to individual pages shall be accomplished in accordance with the change notice guidelines of MIL-STD-962.

4.3.1 Revision to method number. Revision to the document will also reflect a change to the test method number. The revision to the test method shall be denoted by a letter following the original number. For example: The original test method number assigned "Test Method 1, Salt Spray" shall be changed to "Test Method 1A, Salt Spray" when revised.

4.4 Selection of tests. The tests contained herein are intended to be used to provide uniform testing methods and presentation of data for fasteners and fastening. These tests shall be applicable for research, development and procurement of weapons systems, equipments, and fasteners. In selecting the tests that are to be applicable to fasteners in a specific category or type, it is required that those fastener properties be determined that are needed for incorporation of specific fasteners in specific types of weapons systems and equipments and those properties that are essential for fabrication of a satisfactory fastener. After these essential properties have been determined, the appropriate test shall be selected and specified for each property. Care shall be taken to specify only those tests that are needed.

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The following metric test methods are available for selection and reference:

DOD-STD-1312-5	Stress Durability
DOD-STD-1312-7	Vibration
DOD-STD-1312-8	Tensile Strength
DOD-STD-1312-9	Stress Corrosion
DOD-STD-1312-11	Tension Fatigue
DOD-STD-1312-13	Double Shear

4.5 Method of reference. This standard shall be referenced in general and detailed specifications or other documentation for military weapons systems. Tests shall be referenced in the individual fastener specification, fastener specification control drawing, etc., by specifying:

a. The standard number without the revision letter.

b. The standard number with the applicable dash number and revision letter. For example: "the fastener shall be tested for stress durability in accordance with MIL-STD-1312-5A."

c. Test criteria or property values required.

5. DETAIL REQUIREMENTS

5.1 Detail requirements shall be incorporated in the individual test method bookform standard part or section.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Reviewer activities:
Army - AV, AR
Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0142)

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